Fixed Points of Generalized Kaprekar Routines: A Hierarchy

Clay Elmore

2026

1 Introduction

In 1949 Kaprekar noticed that subtracting the ascending digit arrangement of a four-digit number from its descending arrangement, and repeating, sends every input other than the repdigits to 6174 within seven steps [Kaprekar 1955]. At three digits the same procedure reaches 495 [Trigg 1974]; at five it reaches nothing, the orbits settling into cycles instead [Prichett 1981]. These three facts are usually presented as a numerical curiosity of base ten. They are better read as three data points about a single map, and the map is one point in a large family.

Fix a digit length d. Sorting the digits of an integer into descending order and then rearranging them by a permutation produces another integer; the difference of two such rearrangements is a generalized Kaprekar routine. The classical map takes the two rearrangements to be the descending and ascending orders. Allowing the two permutations to be arbitrary gives a family of $d!\,(d!-1)$ maps at each digit length, of which the classical map is one. Within this family 6174 is no longer exceptional: it is one of several numbers fixed by some map at $d = 4$, and the question of which integers are fixed, and how strongly, has a clean answer that the classical map alone obscures.

This paper organizes that answer. Every integer fixed by some generalized Kaprekar routine sits at one of six levels, ordered by how much of the dynamics it controls. At the bottom a fixed point governs nothing — the map collapses almost everything to zero, or fixes the point in name only, with no orbit flowing to it. At the top a single integer is the global attractor of an appropriate map at *every* sufficiently large digit length, the maps linked by an explicit rule for passing from one length to the next. The classical constants occupy the middle: 495 and its relatives are universal but degenerate, fixed by maps that secretly act in fewer dimensions than they appear to; 6174 is the first genuinely full-dimensional universal attractor. Above 6174 the hierarchy splits into two ways of climbing in dimension, and the two best-understood inhabitants of the top level — 60714 and the family generated by 6174 under digit duplication — realize the two ways.

We make the levels precise in Section 3 and populate them in Sections 4 through 7. The two organizing invariants, defined next, are the rank of a map and the rank it exhibits at a given fixed point; the gap between them is what separates degenerate universality from the real thing.

2 Rules and coefficients

Throughout, $d \geq 2$ is a fixed digit length and integers are padded to d digits with leading zeros. For an integer n write $\mathrm{sort}_{\downarrow}(n) = (s_0, s_1, \ldots, s_{d-1})$ for its digits in descending order, $s_0 \geq s_1 \geq \cdots \geq s_{d-1}$.

Definition 2.1. For an ordered pair of permutations $(\pi, \sigma) \in S_d \times S_d$ with $\pi \neq \sigma$, the *rule* $K_{\pi,\sigma}$ acts on n by

$$K_{\pi,\sigma}(n) = \Big| \pi \cdot \mathrm{sort}_{\downarrow}(n) - \sigma \cdot \mathrm{sort}_{\downarrow}(n) \Big|,$$

where $\tau \cdot (s_0, \ldots, s_{d-1})$ denotes the integer $\sum_i 10^{\,d-1-i}\, s_{\tau(i)}$. The classical Kaprekar map is $K_{\mathrm{id},\,\mathrm{rev}}$.

Because a rule reads its input only through the sorted digit vector, it is a function of the digit multiset alone, and the orbit of n depends only on that multiset. Expanding the difference shows that $K_{\pi,\sigma}$ is linear in the sorted digits up to the absolute value:

$$K_{\pi,\sigma}(n) = \Big| \sum_{i=0}^{d-1} c_i\, s_i \Big|, \qquad c_i = 10^{\,d-1-\pi^{-1}(i)} - 10^{\,d-1-\sigma^{-1}(i)},$$

and the coefficients sum to zero. The vector $c = (c_0, \ldots, c_{d-1})$ determines the rule's action on every multiset, so it is the right object to track.

Definition 2.2. The *surviving-variable count* of a rule is $\mathrm{sv}(K) = \#\{i : c_i \neq 0\}$. The rule is *full-variable* when $\mathrm{sv}(K) = d$, that is, when every sorted position contributes to the output.

The classical map is full-variable at $d = 4$, where $c = (999, 90, -90, -999)$, but not at odd lengths: at $d = 3$ the descending-minus-ascending borrow chain forces the middle coefficient to vanish, $c = (99, 0, -99)$, and the map reads only the outer two digits. The same cancellation occurs at every odd length, which is the structural reason the classical map fails to converge at $d = 5$ — there it has rank four, not five, and is not the object one should be asking about.

A second count refines the first at a fixed point. Write F for an integer fixed by K, with sorted digits $(f_0, \ldots, f_{d-1})$.

Definition 2.3. The *effective rank* of K at F is $\mathrm{sv}_F(K) = \#\{i : c_i \neq 0 \text{ and } f_i \neq 0\}$.

Always $\mathrm{sv}_F(K) \leq \mathrm{sv}(K)$, with equality exactly when F has no zero digit in a surviving position. When F carries zeros, a rule can be full-variable as an algebraic object yet act at F through strictly fewer coordinates. This gap is invisible to the fixed-point equation $K(F) = F$ and decisive for everything that follows.

Definition 2.4. A rule K is *universal for F* at length d if every non-repdigit d-digit input reaches F under iteration. We then call F a *universal fixed point* of K.

Repdigits map to zero under every rule and are excluded throughout, as are the near-repdigit degeneracies standard in the literature.

3 The hierarchy

Six conditions on a fixed point F, each strictly stronger than the last, organize the rest of the paper. We state them first and inhabit them afterward.

(L0) Collapse. The rule sends every non-repdigit input to 0. Nothing is fixed except trivially, and there is no dynamics to speak of. Collapse is the floor against which the other levels are measured.

(L1) Bare fixed point. $K(F) = F$, but the basin of F is trivial: no non-repdigit input other than F itself reaches F. Being fixed is a local algebraic accident and carries no dynamical weight. Section 4 exhibits an infinite family of bare fixed points that look substantial and are not.

(L2) Attractor. $K(F) = F$ and a positive proportion of inputs reach F, but not all of them. This is the generic situation and the bulk of the census.

(L3) Degenerate universal. F is universal for K, but $\mathrm{sv}_F(K) < d$: the universality is produced by a rule acting at F through fewer than d coordinates. The classical small-length constants live here.

(L4) Full universal. F is universal for a rule with $\mathrm{sv}_F(K) = d$. Every coordinate participates, and F cannot be obtained by embedding a shorter solution. This is the first level at which universality reflects genuine d-dimensional structure.

(L5) Dimension-transcendent. A single integer, or an explicitly described family, is a full universal fixed point at infinitely many digit lengths, the rules at successive lengths connected by a fixed lifting recipe. Two distinct liftings populate this level, and they meet at 6174.

The levels are nested: L4 implies L3's universality with the rank condition strengthened, L3 implies L2 with the basin filled, and so on down. The content of the paper is that each level is nonempty, that the named classical objects fall where one would hope, and that the top level admits exactly the two constructions described in Sections 6 and 7. The following table fixes the picture; the entries are justified in the sections indicated.

Level	Condition on F	Representative	Section
L0	$K \equiv 0$ off repdigits	—	3
L1	fixed, trivial basin	6174 repeated m times, $m \geq 3$	4
L2	positive but partial basin	generic	4
L3	universal, $\mathrm{sv}_F < d$	45, 450, 495	5
L4	universal, $\mathrm{sv}_F = d$	6174	6
L5	full universal at infinitely many d	60714; the $\{1,4,6,7\}^m$ family	6–7

4 Bare fixed points and attractors

A fixed point need not pull anything toward itself. The cleanest illustration is an infinite family that looks like a generalization of 6174 and behaves like nothing of the sort.

Let $M_m = \{1,4,6,7\}^m$ be the multiset with each of $1,4,6,7$ repeated m times, a length $d = 4m$. Write

$$F_m = \underbrace{6174\,6174 \cdots 6174}_{m} = 6174 \cdot R(m), \qquad R(m) = \sum_{k=0}^{m-1} 10^{4k}.$$

Proposition 4.1. For every $m \geq 1$, F_m is a fixed point of a full-variable rule at $d = 4m$, and F_m has digit multiset M_m.

Proof. Take the rule that, on the sorted input $(7^m, 6^m, 4^m, 1^m)$, places the seven-ranks, six-ranks, four-ranks, and one-ranks into the position classes congruent to $0,1,2,3 \bmod 4$ respectively, and forms the difference against the pairwise-swapped arrangement. The two arrangements read 7641 repeated m times and 1467 repeated m times, so

$$K(F_m) = |7641 \cdot R(m) - 1467 \cdot R(m)| = (7641 - 1467) \cdot R(m) = 6174 \cdot R(m) = F_m.$$

The output is 6174 written m times, with multiset M_m, and the rule is full-variable. □

The identity is the four-digit fact $7641 - 1467 = 6174$ carried across m disjoint blocks by the geometric factor $R(m)$. It is also the entire content: the construction copies the $d = 4$ solution and does nothing else.

Proposition 4.2. The basin of F_m under the rule of Proposition 4.1 is the whole space at $m = 1, 2$ and a strict, erratic fraction of it for $m \geq 3$: the coverage is $100\%, 100\%, 14.9\%, 99.7\%, 9.5\%$ at $m = 1, \ldots, 5$.

At $m = 2$ the rule acts as two copies of the classical four-digit map on the even and odd position classes, which do not interact, so universality is inherited from $d = 4$. At $m \geq 3$ the sorting step mixes the blocks before each subtraction, the copies cease to be independent, and the constructed fixed point is left dynamically isolated — fixed, but reached from almost nothing. F_m is a bare fixed point (L1) for every $m \geq 3$, and the appearance of a pattern generalizing 6174 is an artifact of the algebra, not a feature of the dynamics.

Between the bare points and the universal ones lies the generic case (L2): a fixed point with a positive but incomplete basin. These are abundant. A search of the 63,063,000 arrangements of M_4 at $d = 16$ turns up fixed points at every basin fraction from a handful of multisets up to 99.9%, the overwhelming majority of them partial. Partiality is the default; universality is the exception the rest of the paper is about.

5 Degenerate universals

The smallest universal fixed points are universal for a reason that does not survive inspection: the rule fixing them acts in fewer dimensions than its length advertises.

At $d = 3$ the classical map has coefficient vector $(99, 0, -99)$. Writing the sorted digits as $a \geq b \geq c$, it computes $99(a - c)$ and never reads b. Its output depends on a single gap, and the three-digit routine accordingly drives every non-repdigit input to 495, the fixed point with $a - c = 5$. The numbers 45 and 450 arise the same way from the two other universal rules at $d = 3$; all three have $\mathrm{sv} = 2$.

Definition. Call F a *degenerate universal* (L3) if F is universal for some rule K with $\mathrm{sv}_F(K) < d$.

The three-digit constants are the prototypes. Their universality is real — every input does reach them — but it is the universality of a rank-two map embedded in three coordinates, and the embedding is what lets them recur at larger lengths without acquiring new structure. A degenerate universal at length d extends to $d + 1$ by padding with a coordinate the rule ignores; the extension fixes the same number for the same reason. This stability across length is not transcendence. It is the absence of dimensional content, and Definition 2.3 is what tells the two apart: a degenerate universal has $\mathrm{sv}_F < d$ at every length where it appears, while the objects of the next two sections have $\mathrm{sv}_F = d$.

6 Full universals and transcendence by zero-padding

Imposing $\mathrm{sv}_F = d$ removes the embedded low-rank solutions and leaves the fixed points that use their full length.

Definition. F is a *full universal* (L4) if it is universal for a rule with $\mathrm{sv}_F(K) = d$.

The first is 6174, with $c = (999, 90, -90, -999)$ and both gaps active. Exhaustive classification gives the full universals at small length.

d	full universals	examples
3	0	— (the $d = 3$ universals are degenerate, $\mathrm{sv} = 2$)
4	4	6174, 1746, 2538, 5382
5	33	60714, 60417
6	506	60714 and liftings

The count is not the point; the existence of full universals at $d = 5$, where the classical map has none, is. It shows that the failure at five digits is a failure of the classical rule, not of the digit length.

Among the thirty-three at $d = 5$, one is distinguished by what happens above it. Write 60714 for the fixed point of the rule with coefficient vector $(9900, 9, 90, -9000, -999)$.

Theorem 6.1 (zero-padding transcendence). *An explicit family of full-variable rules, one at each $d \geq 5$ and related by a coefficient-preserving lifting, fixes* 60714 *at every length. Under it* 60714 *is universal at $d = 5$ and $d = 6$, and universal on $A_d \setminus E_d$ at every $d \geq 7$, where E_d is a structurally characterized escape class whose orbits collapse to* 0 *— exactly as the classical map's escape inputs collapse at $d = 3, 4$. The proof is in Appendix D.*

The proof is by a zero-padding lifting argument; we recall only the mechanism here, because the mechanism is what the next section lacks. The lifting appends a zero-sum pair of coefficients at the top two positions, padding F with two trailing zeros. The padded zeros form an absorbing set: once an orbit's sorted form ends in two zeros, the rule acts on the remaining digits as the rule two lengths down, and the question reduces. Two facts close the argument — that this absorbing set is invariant, and that every orbit reaches it in a bounded number of steps — and the second is established by a digit-counting argument that turns on the appended pair producing zeros. The zeros are not incidental. They are the structure that makes the induction run.

By contrast 6174 itself is transcendent only intermittently along this axis: it is universal at $d = 4$ and again at $d = 7$, under a zero-sum lifting, but not at $d = 5$ or $d = 6$. The cross-dimensional behavior of a fixed point is a property of the lifting available to it, and 60714 is the integer for which the zero-padding lifting succeeds uniformly.

7 Transcendence by duplication

Zero-padding is one way to raise the dimension of a fixed point. Duplicating its digits is another, and it leads to the family of Section 4 — but to the universal members, not the bare ones.

Fix the multiset $M_m = \{1, 4, 6, 7\}^m$ at $d = 4m$. Section 4 produced one fixed point of this multiset, F_m, and showed it bare for $m \geq 3$. The multiset carries others. Among the rules fixing some arrangement of M_m, certain ones are universal, and they are not the rule of Proposition 4.1.

Computational result 7.1. *Full universal fixed points of multiset M_m exist at $m = 1, 2, 3, 4, 5, 6$. For $m \geq 3$ no universal arrangement decomposes into blocks; the universal arrangements are necessarily scrambled.*

m	d	universals	a verified witness
1	4	2	6174
2	8	465	61746174
3	12	46	666141417774
4	16	≥ 313	6614617774614174
5	20	≥ 1	14617461774617461746
6	24	≥ 1	666174141466617777741414

The counts at $m = 1, 2$ are exhaustive over all full-variable rules; at $m = 3$ exhaustive within the pair-symmetric family; at $m = 4$ a verified lower bound from a structured search covering roughly a third of the rule space; at $m = 5, 6$ the witnesses were found by targeted search. Every witness in the table was checked to be a genuine full-variable fixed point. The basins at $m = 1, 3, 4, 6$ were confirmed to be the entire admissible input set — $\binom{d+9}{9} - 10$ digit multisets — by direct enumeration; at $m = 6$ this is all 38,567,090 admissible inputs at $d = 24$. At $m = 5$ the admissible set exceeds ten million multisets and full enumeration is impractical, so universality there was checked on a reduced alphabet-proxy rather than the complete input set. The case $m = 2$ gives 61746174 and four hundred sixty-four others. From $m = 3$ on, the clean block-structured candidate F_m is not among the universals — its basin is the 14.9% of Proposition 4.2 — and the universal arrangements have no block decomposition at all. Whether the family continues was the question we could not settle by construction, and the reason is worth stating precisely, because the obstruction is sharper than "the search got hard."

Restrict to the rules natural to this multiset — the pair-symmetric rules, in which the swapped arrangement pairs the outer digit blocks and the inner digit blocks — and to those whose coefficient partial sums are nonnegative, a monotonicity that the classical four-digit rule already satisfies. For these,

Lemma 7.2. *A rule is universal if and only if its dynamics has a unique fixed point and no cycle of period at least two.*

This is a finite-state fact: in a finite space every orbit ends at a fixed point or a cycle, so the absence of competing fixed points and of cycles is exactly global convergence. It reduces universality to the exclusion of cycles, and the cycles can be read off. Every cycle of every such rule lives outside the four-digit alphabet — its members carry digits other than $1, 4, 6, 7$ — and at each length the cycles fall into finitely many families. The difficulty is that the families do not stabilize: the cycles obstructing universality at $d = 20$ have periods and digit-profiles absent at $d = 12$ and $d = 16$, and the partition data that characterizes the universal rules at $m = 3$ and $m = 4$ fails to characterize them at $m = 5$. There is no description of the universal rules uniform in m.

What does hold uniformly is abundance. The number of universal rules, estimated by sampling against the exact count at $m = 3$, grows as roughly $3 \times 10^3,\ 2 \times 10^7,\ 3 \times 10^8$ at $m = 3, 4, 5$; the universal *fraction* falls only because the rule count grows faster. The witnesses are not becoming scarce. They are becoming unstructured.

Conjecture 7.3. *Multiset* $\{1,4,6,7\}^m$ *admits a full universal fixed point at every* $m \geq 1$*.*

The evidence is the growth of Result 7.1 and the witness count. What is missing is any of the structures that prove such statements — a uniform construction, an inductive lifting, an invariant separating universal rules from the rest — and the next section locates why none has been found.

8 The base case has no conceptual certificate

The multiplicity conjecture asks for convergence at every length in a family whose first member is the classical four-digit map. One would prove it by exhibiting a mechanism at $d = 4$ and lifting it. The obstruction is that the four-digit map has no such mechanism to lift.

Reduce the classical map to its gaps. For sorted digits $a \geq b \geq c \geq d$, the map computes $999(a-d)+90(b-c)$ and so depends only on $p = a-d$ and $q = b-c$, with $1 \leq p \leq 9$ and $0 \leq q \leq p$. This is a map on fifty-four states, with unique fixed point $(p,q) = (6,2)$, and convergence to 6174 is convergence of this finite system. The question is whether the convergence has a certificate simpler than tracing the states.

Proposition 8.1. *The fifty-four-state gap system admits no polynomial Lyapunov function of degree at most four: there is no polynomial* Φ *of degree* ≤ 4 *with* $\Phi(\textit{next state}) < \Phi(\textit{state})$ *off the fixed point.*

The statement is a finite linear feasibility problem and it is infeasible at degrees two, three, and four. A Lyapunov function does exist at degree nine, but only because nine is the degree at which a polynomial can interpolate fifty-four prescribed values, and the values it interpolates are the number of steps each state takes to reach $(6,2)$ — the reaching time, which decreases by one along every orbit by definition and certifies convergence on any finite acyclic system whatever. It is not a reason. It is the dynamics rewritten as a polynomial.

So the convergence of the classical map, the seventy-five-year-old fact underneath everything here, has no low-complexity certificate. It is established by enumeration or by an interpolation of its own reaching time, and neither generalizes. The zero-padding transcendence of Section 6 evades this because it does not lift a mechanism through the dynamics; it builds an absorbing set out of zeros and reduces the length directly. The multiplicity chain has no zeros and no absorbing set, so it has no route around the base case. A proof of Conjecture 7.3, if there is one, will not come from the dynamics of any single length. It will come either from a genuine structural reason for the four-digit convergence — which would be a result in its own right — or from an existence argument that counts the universal rules without describing them, of the kind the witness-count growth invites.

A Proof of the universality criterion

Let T be the map $n \mapsto \mathrm{sort}_{\downarrow}|K(n)|$ acting on the finite set X of non-repdigit d-digit multisets, and let $F \in X$ be a fixed point, $T(F) = F$. We prove Lemma 7.2 in the form: F is universal — meaning $T^k(x) = F$ for some k, for every $x \in X$ — if and only if F is the only fixed point of T and T has no periodic orbit of period at least two. The argument uses only finiteness of X and applies to every rule, not merely the pair-symmetric monotone ones of Section 7.

Suppose first that F is universal. If $G \neq F$ were a second fixed point, then $T^k(G) = G \neq F$ for all k, so G never reaches F, contradicting universality; hence F is the unique fixed point. If C were a periodic orbit of period $p \geq 2$, then for any $x \in C$ the iterates $T^k(x)$ cycle through C and return to x, never leaving C; since $F \notin C$ (a fixed point is its own orbit, of period one), no point of C reaches F, again contradicting universality. So neither a second fixed point nor a long cycle can exist.

Conversely, suppose F is the unique fixed point and T has no periodic orbit of period ≥ 2. Fix any $x \in X$. Because X is finite, the forward orbit $x, T(x), T^2(x), \ldots$ cannot be injective, so $T^a(x) = T^{a+p}(x)$ for some $a \geq 0$ and $p \geq 1$. The set $\{T^a(x), T^{a+1}(x), \ldots, T^{a+p-1}(x)\}$ is then a periodic orbit; by hypothesis it has period one, so $T^a(x)$ is a fixed point, and by uniqueness $T^a(x) = F$. Thus x reaches F, and F is universal. ∎

The two conditions are independent — a rule may have a unique fixed point yet trap part of its space in a cycle, and the d=12 interleaved monotone rule cited in Section 7 has two competing fixed points with no long cycle — so both must be checked. Each is a finite search over X, which is what makes universality decidable in principle; the open problem of Section 7 is that one of the two searches, the exclusion of cycles, has no answer uniform in the length.

B The four-digit gap system admits no low-degree certificate

In the gap coordinates of Section 8 the classical four-digit map is a self-map G of the fifty-four-element set $\Delta = \{(p, q) : 1 \leq p \leq 9,\ 0 \leq q \leq p\}$ with unique fixed point $(6, 2)$. A polynomial Lyapunov function is a $\Phi(p, q)$ that strictly decreases along every nonstationary step,

$$\Phi(G(p, q)) < \Phi(p, q) \quad \text{for all } (p, q) \in \Delta \setminus \{(6, 2)\}. \tag{$*$}$$

Fix a degree bound D and expand $\Phi = \sum_{i+j \leq D} w_{ij}\, p^i q^j$. Each of the fifty-three inequalities in $(*)$ is linear in the coefficient vector w: writing $\mathbf{m}(p, q)$ for the vector of monomials $p^i q^j$ evaluated at (p, q), the inequality at state (p, q) reads $w \cdot (\mathbf{m}(G(p, q)) - \mathbf{m}(p, q)) \leq -1$, where the right-hand margin -1 fixes the scale, any strict certificate being rescalable to unit margin. The existence of a degree-$\leq D$ Lyapunov function is therefore exactly the feasibility of a linear program in the $\binom{D+2}{2}$ unknowns w_{ij} subject to these fifty-three constraints, with the exact integer transition data of G as input.

Proposition 8.1. *This linear program is infeasible for $D = 2, 3, 4$; no polynomial Lyapunov function of degree at most four certifies $(*)$.*

Infeasibility of a rational linear program is itself a finite, exact certificate — a nonnegative combination of the constraints summing to the contradiction $0 \leq -1$ exists by Farkas's lemma, and is verifiable in integer arithmetic independent of any floating-point solver. At $D = 9$ the program becomes feasible, but vacuously: the monomials of degree ≤ 9 number $\binom{11}{2} = 55 > 54 = |\Delta|$, so they span the full space of real functions on Δ, and one may take Φ to interpolate the reaching time $\rho(p, q) = \min\{k : G^k(p, q) = (6, 2)\}$. This ρ decreases by exactly one along every step by its definition and so satisfies $(*)$ — but it certifies convergence of *any* finite map whose functional graph is a tree rooted at its fixed point, encoding the conclusion rather than explaining it. The content of Proposition 8.1 is the gap between degree four and this interpolation threshold: no certificate of bounded complexity below it exists, which is why no mechanism at $d = 4$ presents itself for lifting.

C Full universals at length $d \leq 6$

The census of Section 6 rests on an exhaustive classification. At each length $d \leq 6$ every full-variable rule — every pair (π, σ) whose coefficient vector has no zero entry — is iterated from every admissible digit multiset, and a fixed point is recorded as universal when its basin is all of A_d. A rule depends only on the sorted digit vector, so the search runs over multisets rather than integers, and that is what keeps $d = 6$ in reach. The number of full-variable rules is $d!\,D_d$ with D_d the derangement count (216, 5,280, 190,800 at $d = 4, 5, 6$), and the number of admissible multisets is $\binom{d+9}{9}$ less the ten repdigits and the ninety near-repdigits (615, 1,902, 4,905 at $d = 4, 5, 6$).

Theorem C.1. *The numbers of universal full-variable fixed points at $d = 3, 4, 5, 6$ are $0, 4, 33, 506$. At $d = 4$ they are $1746, 2538, 5382, 6174$, splitting into the anagram clusters $\{1, 4, 6, 7\}$ ($6174, 1746$) and $\{2, 3, 5, 8\}$ ($2538, 5382$); all four have digit sum $\equiv 0 \pmod 9$.*

Proof. Exhaustive enumeration. At $d = 3$ the twelve full-variable rules each leave some admissible input in a cycle or at a non-universal fixed point, so none has full basin — the genuine three-digit universals $45, 450, 495$ are not full-variable, their rules reading only two of the three positions (sv $= 2$). At $d = 4, 5, 6$ each full-variable rule is iterated against every admissible multiset and basin coverage of A_d is checked directly; the universal fixed points are those with full basin. The counts are as stated. ■

Theorem C.2 (cross-dimensional selection). *Of the 33 universal full-variable fixed points at $d = 5$, exactly one — 60714 — is again a universal full-variable fixed point under some rule at $d = 6$. The other 32 are dimension-locked: no full-variable rule at $d = 6$ is universal for any of them.*

Proof. For each of the 33 fixed points F, enumerate the full-variable rules at $d = 6$ that fix the zero-padded F and test each for universality. For 32 of them either no full-variable rule fixes the padded form (algebraic obstruction) or every rule that does has basin below the whole of A_6 (dynamic obstruction). For $F = 60714$ exactly two rules at $d = 6$ are universal, and they form a sign-flip pair. ■

This selection is what singles out 60714: it is chosen not for its digits but as the unique survivor of the $d = 5 \to d = 6$ test, and the next appendix shows the survivor extends to every length.

D The zero-padding transcendence of 60714

This appendix proves Theorem 6.1. Throughout, F is a universal fixed point at a native length d_F with a universal rule of coefficient vector $c^{(d_F)}$, and $N = \{i : f_i \neq 0\}$ is the set of nonzero-digit positions of F's sorted form.

Definition D.1. A rule at length $d > d_F$ with coefficient vector $\tilde{c}$ is a *coefficient-preserving lifting* of the native rule if $\tilde{c}_i = c_i^{(d_F)}$ for every $i \in N$ — the coefficients at F's nonzero-digit positions are copied verbatim.

Proposition D.2 (fixed-point preservation). *A coefficient-preserving lifting fixes F zero-padded to d digits.*

Proof. The sorted form of the padded F is f followed by $d - d_F$ zeros. In $|\sum_i \tilde{c}_i f_i|$ every term with $i \notin N$ multiplies a zero digit, so only the N-terms survive, and there $\tilde{c}_i = c_i^{(d_F)}$. The sum is therefore $|\sum_{i \in N} c_i^{(d_F)} f_i| = K^{(d_F)}(F) = F$. ■

The fixed-point equation is thus free: all content of a lifting lives at F's zero-digit positions, where it cannot disturb $K(F) = F$. What is not free is universality, and that is the rest of the appendix.

The two ladders. At $d_F = 5$ the native rule for 60714 has $c^{(5)} = (9900, 9, 90, -9000, -999)$ and sorted form $(7, 6, 4, 1, 0)$, so $N = \{0, 1, 2, 3\}$ with the one zero at position 4. The lifting appends, two positions at a time, a *zero-sum pair* $(+9 \cdot 10^d,\ -9 \cdot 10^d)$ at the two new top positions, padding F with two further zeros. This yields an odd ladder ($d = 5, 7, 9, \ldots$) rooted at $c^{(5)}$ and an even ladder ($d = 6, 8, \ldots$) rooted at the split lifting $c^{(6)} = (9900, 9, 90, -9000, 99000, -99999)$, where the native -999 is carried by two coefficients $99000 + (-99999) = -999$ at the two zero positions. By Proposition D.2 every rule on both ladders fixes 60714.

Definition D.3. The *tail-two-zeros set* $T_d \subseteq A_d$ is the admissible inputs whose sorted form ends in two zeros, $s_{d-1} = s_{d-2} = 0$.

Lemma D.4 (structural closure). *For $d \geq 7$ on either ladder and every $n \in T_d$ with sorted form $(s_0, \ldots, s_{d-3}, 0, 0)$,*

$$K^{(d)}(n) = K^{(d-2)}((s_0, \ldots, s_{d-3})) \in T_d.$$

Proof. The appended pair occupies the last two positions, which carry $s_{d-2} = s_{d-1} = 0$, so it contributes nothing and $K^{(d)}(n) = |\sum_{i=0}^{d-3} c_i^{(d-2)} s_i| = K^{(d-2)}((s_0, \ldots, s_{d-3}))$. The output of a rule at $d - 2$ is below 10^{d-2}, hence has at most $d - 2$ digits; padded to d digits it carries at least two leading zeros, which sort to the tail, so the image lies in T_d. ■

The proof uses only the zero-sum pair structure and holds for any such lifting. With closure in hand the theorem reduces to one dynamical fact.

Lemma D.5 (bounded reaching time). *For $d \geq 7$ on either ladder, every admissible $n \in A_d \backslash E_d$ reaches T_d in a bounded number of steps, with E_d the escape class defined below.*

Granting Lemmas D.4 and D.5, Theorem 6.1 follows by induction in steps of two along each ladder. The roots $d = 5, 6$ are universal by Appendix C. For $d \geq 7$, an orbit reaches T_d by Lemma D.5, and there it acts as the rule at $d - 2$ by Lemma D.4, which reaches 60714 by induction.

Reaching time: the algebra. Split the output into the *core* — the five native positions — plus the appended pairs, writing $\mathrm{core}(s) = 9900s_0 + 9s_1 + 90s_2 - 9000s_3 - 999s_4$.

Lemma D.6 (core bounds). *For every sorted-descending s, $0 \leq \mathrm{core}(s) \leq 89{,}991 < 10^5$.*

Proof. Upper bound: the negative terms are ≤ 0, so $\mathrm{core}(s) \leq 9900 \cdot 9 + 9 \cdot 9 + 90 \cdot 9 = 89{,}991$. Lower bound, by cases on (s_3, s_4). If $s_3 = s_4 = 0$, the core is a sum of non-negative terms. If $s_3 \geq 1,\ s_4 = 0$, then $\mathrm{core}(s) = 9000(s_0 - s_3) + 900s_0 + 9s_1 + 90s_2 \geq 0$. If $s_3, s_4 \geq 1$, apply $s_2 \geq s_3$, then $s_1 \geq s_4$, then $s_4 \leq s_3 \leq s_0$:

$$\mathrm{core}(s) \geq 9900s_0 + 90s_3 + 9s_4 - 9000s_3 - 999s_4 \geq 9900s_0 - 8910s_3 - 990s_4 \geq 9900(s_0 - s_3) \geq 0.$$

The remaining case $s_3 = 0,\ s_4 \geq 1$ is excluded by sorting. ■

Each appended pair k contributes $p_k \delta_k$ with $p_k = 9 \cdot 10^{3+2k}$ and $\delta_k = s_{3+2k} - s_{4+2k} \in \{0, \ldots, 9\}$.

Lemma D.7 (cliff-sum bound). $\sum_{k=1}^{M} \delta_k \leq 9$, *where M is the number of appended pairs.*

Proof. The δ_k are disjoint non-negative adjacent differences within the sorted sequence, so their sum is at most the telescoping total $s_3 - s_{d-1} \leq 9$. ■

By Lemma D.6 the core is non-negative and below 10^5, so the argument of the absolute value is non-negative and

$$K^{(d)}(n) = \mathrm{core}(s) + \sum_{k=1}^{M} 9 \cdot 10^{3+2k}\, \delta_k.$$

Read off the decimal blocks: a pair with $\delta_k = 0$ writes two zero digits, $\delta_k = 1$ writes one, $\delta_k \geq 2$ none. Since $\sum_k \delta_k \leq 9$, at most nine of the M blocks fail to contribute a zero, so the output carries at least $2M - 9$ zero digits in the block region. On the odd ladder at $d \geq 15$ (so $M \geq 5$) this leaves at least one block-region zero, which together with a zero forced by the sub-10^5 core gives the two trailing zeros of T_d — one-step closure. The even ladder gives the same conclusion within two steps at $d \geq 16$. The finitely many lower lengths are closed by enumeration.

Proposition D.8 (finite-state verification). *For each $d \in \{7, 8, \ldots, 16\}$ and each ladder's rule, every admissible input reaches T_d within at most eight steps.*

Proof. Direct enumeration over all admissible multisets at each d — from 11,340 at $d = 7$ to 2,042,875 at $d = 16$, about twenty-five million in all — iterating until two trailing zeros appear. The maximum step count is eight at $d = 7$ and falls to one by $d = 15$, with no exceptions. ■

Proposition D.8 and the algebraic argument together prove Lemma D.5.

The escape class. The reduction is exact except on inputs the rule sends to 0. The *block-aligned* multisets — constant on the native block and on each appended pair — die in one step, since the native coefficients sum to zero and each pair sees equal digits; call this set B_d. The escape class E_d is the backward orbit of B_d under the rule. It is empty at $d = 5, 6$, and at $d \geq 7$ its step-one part $B_d \cap A_d$ has the closed-form size $\binom{9+k}{k} - 10$ with k the number of blocks (45 at $d = 7, 8$, matching exhaustive enumeration). Every E_d orbit reaches 0, just as the classical map sends its own block-aligned inputs — the repdigits at $d = 3$, the multiples of 1111 at $d = 4$ — to 0 rather than to the constant. What is proven is universal convergence on $A_d \setminus E_d$; a bound on $|E_d|$ uniform in d is not, and is the one part of the picture left open.

Conclusion. Lemmas D.4–D.7 and Proposition D.8 establish, with no conjectural step, that 60714 is universal at $d = 5, 6$ and universal on $A_d \setminus E_d$ at every $d \geq 7$ under the lifting family. This is Theorem 6.1. The zeros do the work: they are the absorbing structure on which the length drops by two. The multiplicity chain of Section 7, having no zeros, has nothing to absorb into — which is the contrast Section 8 turns on.

Acknowledgments

The computational census, the structured rule searches, the adversarial verification of every dynamical claim, and the development and testing of the structural conjectures were carried out in collaboration with Anthropic's Claude, used as a research instrument for code generation, large-scale enumeration, and iterative hypothesis testing. The mathematical direction, the choice of questions, and responsibility for the results are the author's. Several intermediate claims were retracted after verification contradicted them; the corrections are reflected in the final statements.

References

[Kaprekar 1955] D. R. Kaprekar, "An interesting property of the number 6174," *Scripta Mathematica*, vol. 15, pp. 244–245, 1955.

[Trigg 1974] C. W. Trigg, "All three-digit integers lead to 495," *The Mathematics Teacher*, vol. 67, pp. 41–45, 1974.

[Prichett 1981] G. D. Prichett, A. L. Ludington, and J. F. Lapenta, “The determination of all decadic Kaprekar constants,” *The Fibonacci Quarterly*, vol. 19, pp. 45–52, 1981. (Precursor: G. D. Prichett, “Kaprekar’s routine with five-digit integers,” *The Fibonacci Quarterly*, vol. 17, pp. 154–159, 1979.)